

# Linear Algebra (MA327)

*University of Campinas*

Fall 2016

## 1 Description

This course offers an introduction to the fundamental concepts of Linear Algebra, which is a common language used in different areas of human knowledge, such as physics, biology, economics, computer science, and engineering. We provide numerous examples illustrating concrete applications of the concepts studied.

## 2 Program

Linear systems. Real vector spaces. Subspaces. Bases and dimensions. Linear dependence and independence. Finite dimensional spaces. Linear transformations and their matrix representations. Kernel and image. Direct sum of subspaces. Projections. Eigenvalues and eigenvectors. Inner product. Orthogonality. Gram-Schmidt orthogonalization process. Cauchy-Schwarz inequality. Adjoint of a linear transformation. Symmetric and orthogonal matrices. Diagonalization.

## Bibliography

- [1] Kostrikin, A. I., & Manin, Y. I. (1997). *Linear Algebra*. CRC Press.
- [2] Hoffman, K. M., & Kunze, R. (1971). *Linear Algebra*. 2nd ed., Pearson.
- [3] Lang, S. (1987). *Linear Algebra*, Springer Verlag.
- [4] Halmos, P. (1950). *Finite-Dimensional Vector Spaces*, Springer Verlag.

CB06 / Tuesdays & Thursdays - 2:00pm-4:00pm

Teaching assistant: Heitor Baldo ([hbaldo@ime.unicamp.br](mailto:hbaldo@ime.unicamp.br))